**PRAKTIKUM SISTEM OPERASI**

**MODUL 4**

**MEMBACA SORCE CODE XINU**

Oleh:

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**PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK**

**DIREKTORAT KAMPUS PURWOKERTO**

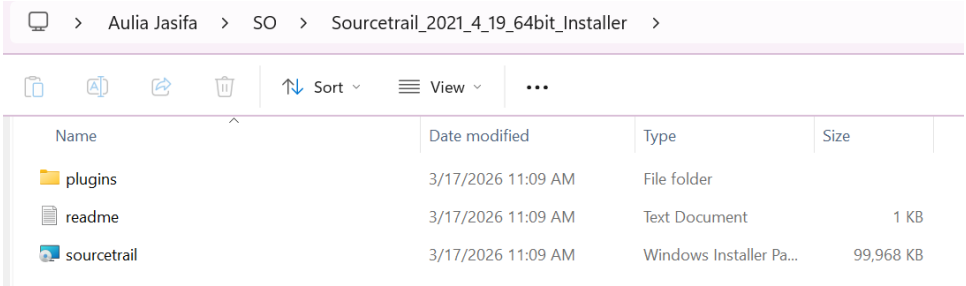
**UNIVERSITAS TELKOM**

**2026**

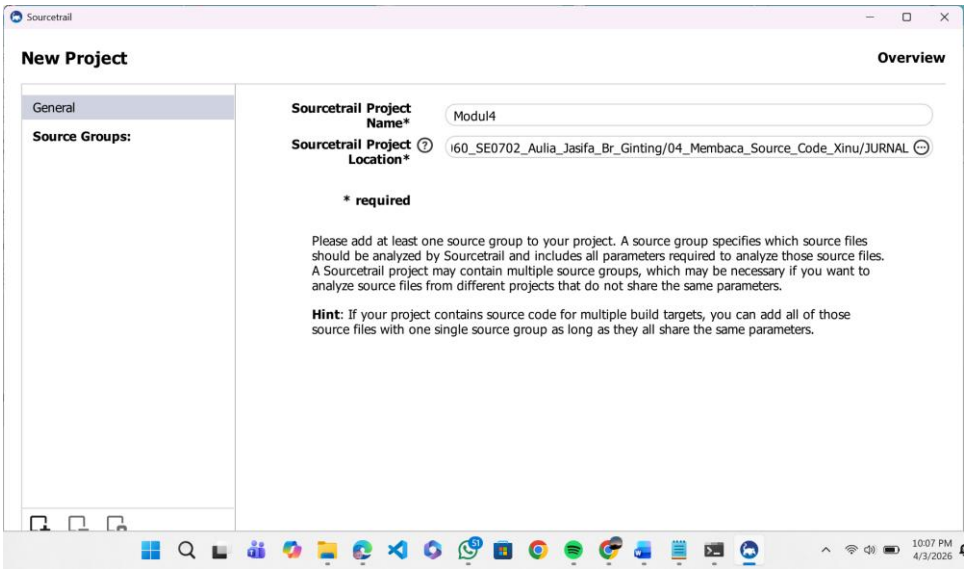
# JURNAL

1. **[10 Poin]** Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?
   Hint: baca kembali modul-modul sebelumnya
   Jawaban: **Nama image**: biasanya bernama xinu.elf atau xinu.bin (tergantung konfigurasi build). **Lokasi file**: berada di folder: xinu/compile/. **Ukuran file**: bervariasi (± ratusan KB, tergantung build & konfigurasi)

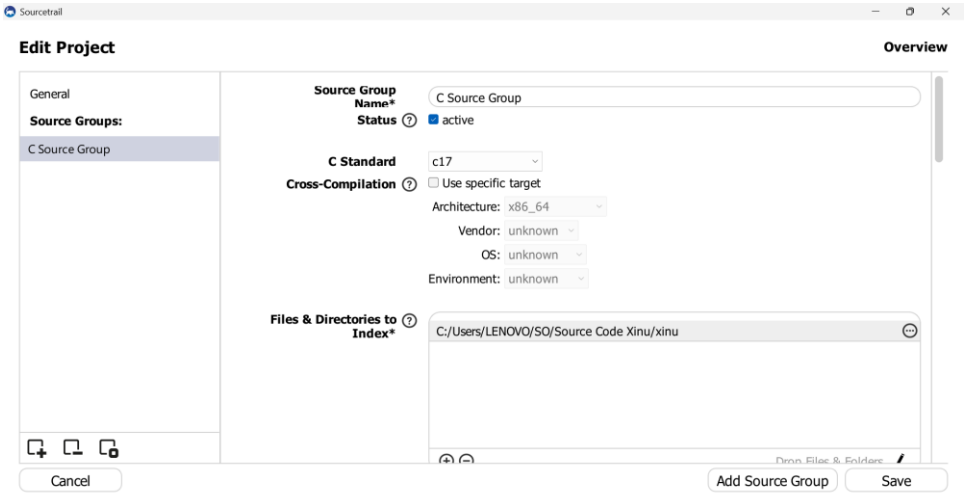
2. Membaca source code Xinu
   a. Cek aplikasi bernama Sourcetrail di PC. Jika belum ada, download SourceTrail pada (DOWNLOAD SOURCETRAIL). SourceTrail adalah software untuk mengeksplorasi source code. Programmer yang bagus lebih banyak membaca kode daripada menulis kode

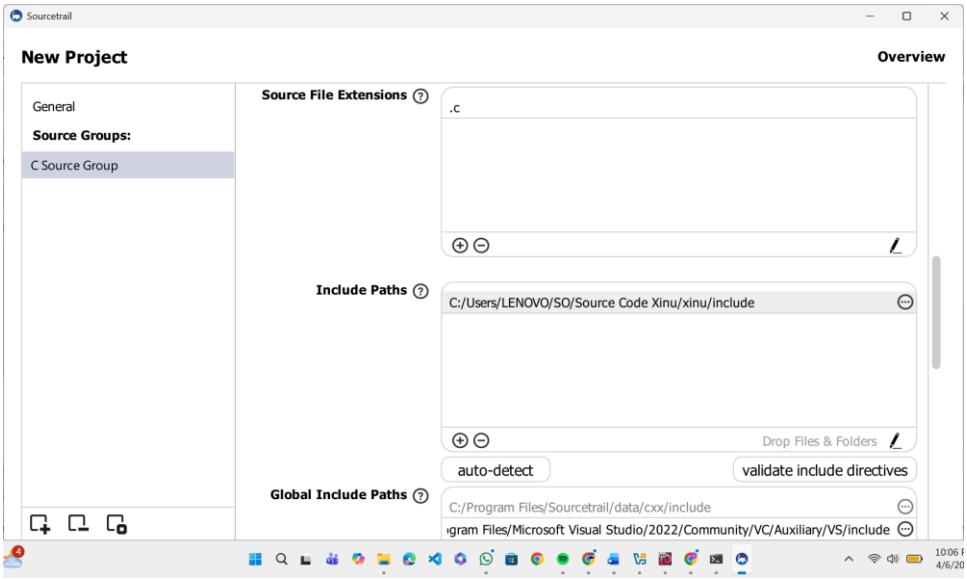



   b. Jalankan SourceTrail
   c. Project → New Project

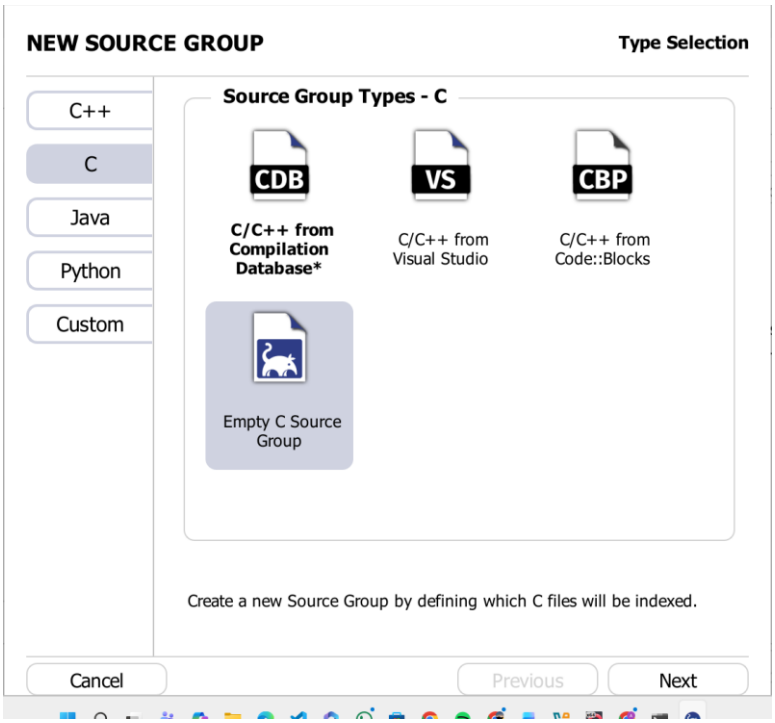


d. Isi nama project xinu dan pilih lokasi project di manapun

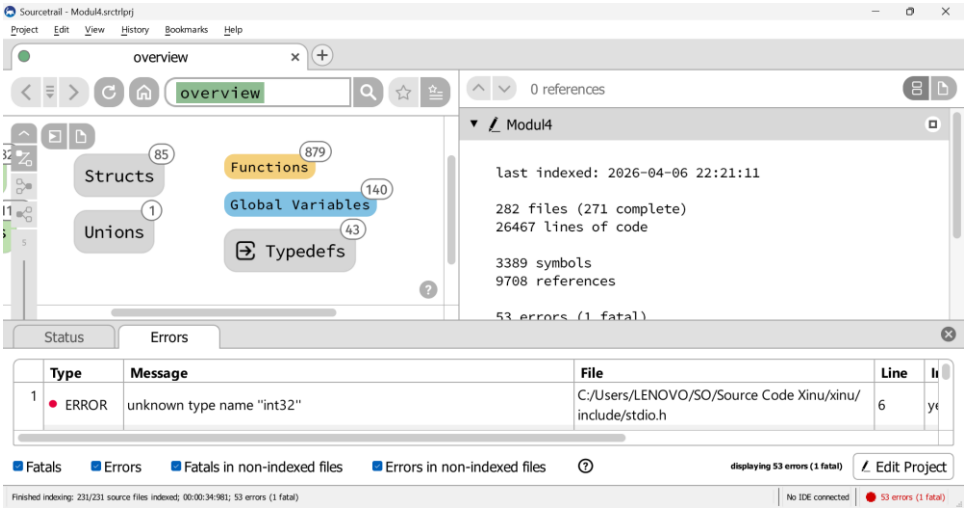


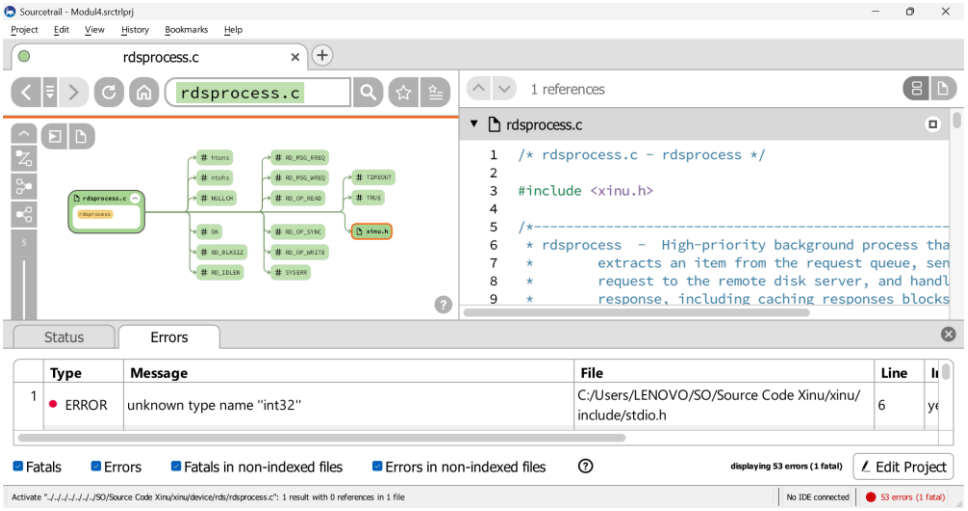



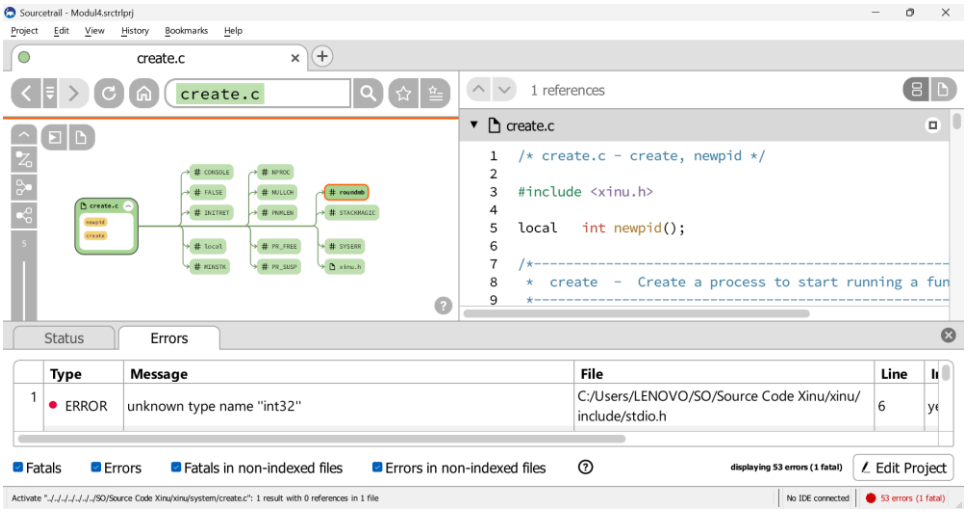

e. Add Source Groups, pilih C, lalu pilih Empty C Source Group




f. File & Directories to Index: masukkan semua folder Xinu (yang sebelumnya telah di download)
g. Include Paths: …/xinu/include
h. Create
i. Silahkan eksplorasi source code Xinu




Sourcetrail - Modul4.srctrlprj
Project Edit View History Bookmarks Help
rdsprocess.c
1 references
/* rdsprocess.c - rdsprocess */
#include <xinu.h>
/*----------
* rdsprocess - High-priority background process tha
* extracts an item from the request queue, sen
* request to the remote disk server, and handl
* response, including caching responses blocks
Status
Errors
Type
Message
File
Line
1
ERROR
unknown type name "int32"
C:/Users/LENOVO/SO/Source Code Xinu/xinu/include/stdio.h
6
Fatals
Errors
Fatals in non-indexed files
Errors in non-indexed files
displaying 53 errors (1 fatal)
Edit Project
No IDE connected
53 errors (1 fatal)


Sourcetrail - Modul4.srctrlprj
Project Edit View History Bookmarks Help
create.c
1 references
/* create.c - create, newpid */
#include <xinu.h>
local int newpid();
/*----------
* create - Create a process to start running a fun
*----------
Status
Errors
Type
Message
File
Line
1
ERROR
unknown type name "int32"
C:/Users/LENOVO/SO/Source Code Xinu/xinu/include/stdio.h
6
Fatals
Errors
Fatals in non-indexed files
Errors in non-indexed files
displaying 53 errors (1 fatal)
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No IDE connected
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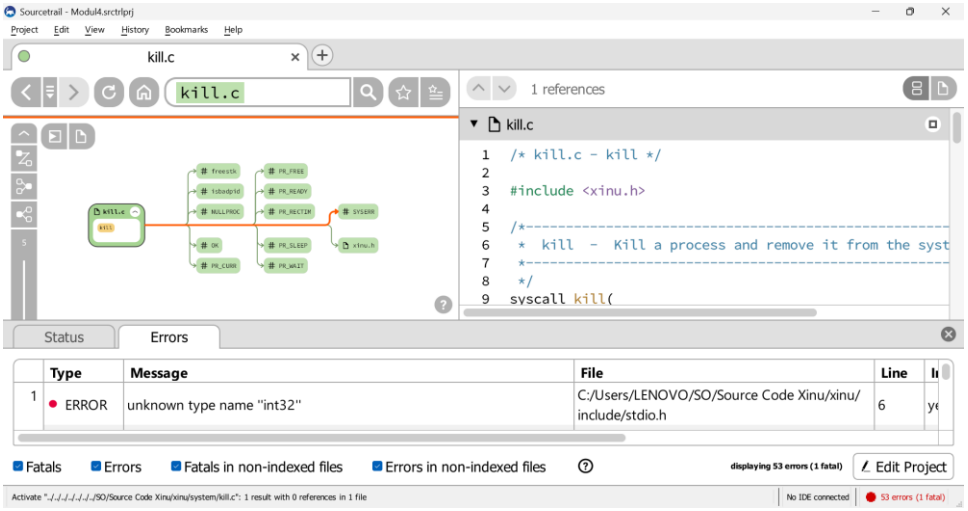
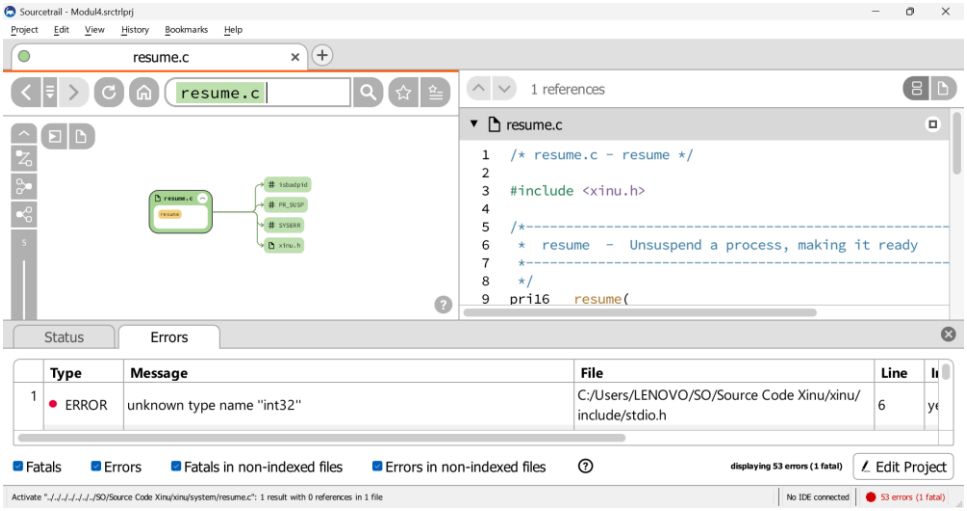
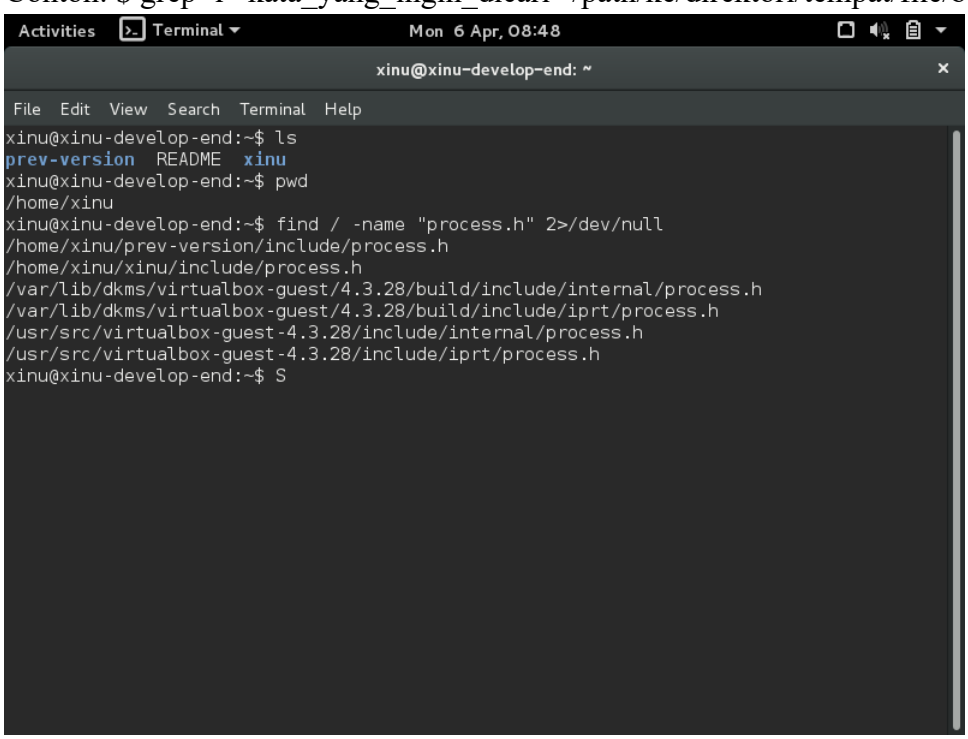

Sourcetrail - Modul4.srctrlprj
Project Edit View History Bookmarks Help
kill.c
1 references
/* kill.c - kill */
#include <xinu.h>
/*----------
* kill - Kill a process and remove it from the syst
*----------
*/
syscall kill(
Status
Errors
Type
Message
File
Line
1
ERROR
unknown type name "int32"
C:/Users/LENOVO/SO/Source Code Xinu/xinu/include/stdio.h
6
Fatals
Errors
Fatals in non-indexed files
Errors in non-indexed files
displaying 53 errors (1 fatal)
Edit Project
No IDE connected
53 errors (1 fatal)




3. **[10 Poin]** Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut? Hint: file berektensi .h

   Pada Linux bisa digunakan perintah "grep"

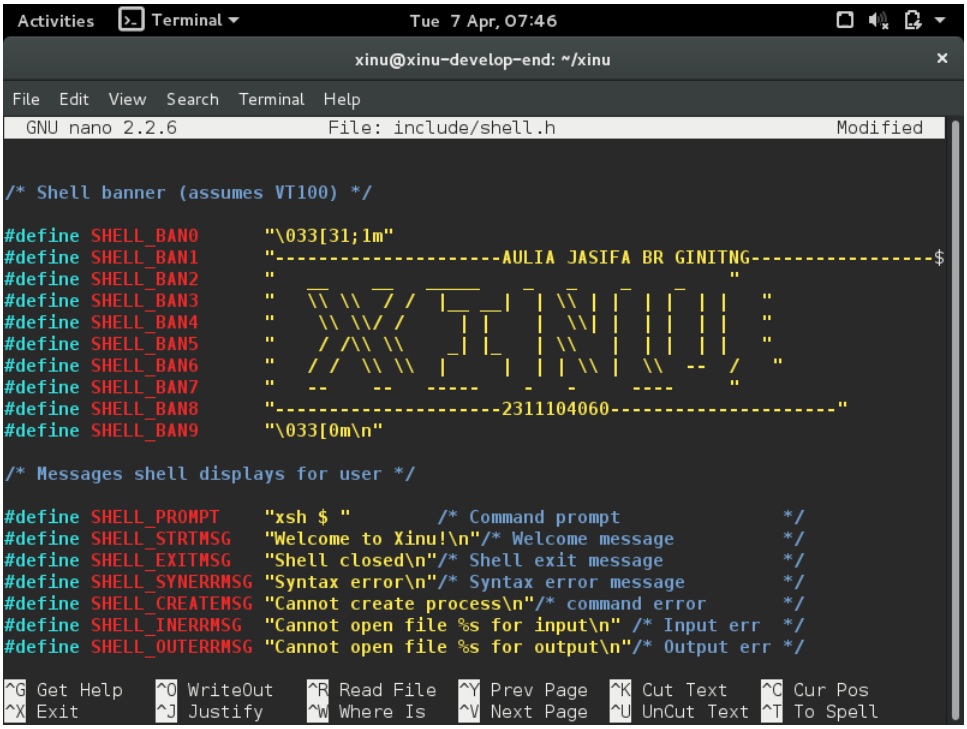
   Contoh: $ grep -r "kata_yang_ingin_dicari" /path/ke/direktori/tempat/file/berada



```
Activities  Terminal  Mon 6 Apr, 08:53
xinu@xinu-develop-end: ~
File  Edit  View  Search  Terminal  Help
xinu@xinu-develop-end:~$ ls
prev-version  README  xinu
xinu@xinu-develop-end:~$ pwd
/home/xinu
xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/include
/home/xinu/xinu/include/shell.h:/* Shell banner (assumes VT100) */
xinu@xinu-develop-end:~$ grep -r "banner" /home/xinu/xinu/shell/
/home/xinu/xinu/shell/shell.c:  /* Print shell banner and startup message */
xinu@xinu-develop-end:~$ grep -r "struct procent" /home/xinu/xinu/include/
/home/xinu/xinu/include/process.h:struct procent {              /* Entry in the process
 table          */
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC           8
#endif

/* Process state constants */
```

```
Activities  Terminal  Mon 6 Apr, 08:53
xinu@xinu-develop-end: ~
File  Edit  View  Search  Terminal  Help
/* Process state constants */

#define PR_FREE         0       /* Process table entry is unused        */
#define PR_CURR         1       /* Process is currently running         */
#define PR_READY        2       /* Process is on ready queue            */
#define PR_RECV         3       /* Process waiting for message          */
#define PR_SLEEP        4       /* Process is sleeping                  */
#define PR_SUSP         5       /* Process is suspended                 */
#define PR_WAIT         6       /* Process is on semaphore queue        */
#define PR_RECTIM       7       /* Process is receiving with timeout    */

/* Miscellaneous process definitions */

#define PNMLEN          16      /* Length of process "name"             */
#define NULLPROC        0       /* ID of the null process               */

/* Process initialization constants */

#define INITSTK         65536   /* Initial process stack size           */
#define INITPRIO        20      /* Initial process priority             */
#define INITRET         userret /* Address to which process returns     */

/* Inline code to check process ID (assumes interrupts are disabled)    */

#define isbadpid(x)     ( ((pid32)(x) < 0) || \
                          ((pid32)(x) >= NPROC) || \
                          (proctab[(x)].prstate == PR_FREE))
```

4. **[80 poin]** Mengubah welcome banner pada Xinu
   a. Carilah file yang menyimpan banner Xinu!
      Hint: file berekstensi .h pada direktori xinu/include
   b. Carilah file yang menampilkan banner Xinu!
      Hint: file berektensi .c pada direktori xinu/shell
   c. Ubahlah welcome banner Xinu sehingga menjadi seperti ini:
      Tambahkan NAMA dan NIM masing-masing, kemudian ubah banner Xinu sesuai dengan selera Anda




   d. Setelah melakukan perubahan lakukan perintah berikut ini pada terminal Linux
      - $ cd xinu/compile
      - $ make clean
      - $ make
      - $ sudo minicom
      - Jalankan vm backend

```
Activities   Terminal   Mon 13 Apr, 20:48
xinu@xinu-develop-end: ~/xinu/compile
File  Edit  View  Search  Terminal  Help
   4447448 bytes of free memory.  Free list:
           [0x0015E320 to 0x0009FFF7]
           [0x00100000 to 0x005FBFFF]
    115657 bytes of Xinu code.
           [0x00100000 to 0x0011C3C8]
    148488 bytes of data.
           [0x0011FEC0 to 0x001442C7]

Obtained IP address  10.0.3.15   (0x0a00030f)



---------------------AULIA JASIFA BR GINITNG---------------------
   __   __  _____   _  _   _   _
   \ \  / /  |__ __|  | \ | |  | |  | |
    \ \/ /     | |    |  \| |  | |  | |
    / /\ \    _| |_   | \   |  | |  | |
   / /  \ \  |     |  | | \ |  \  --  /
   --    --  -----    -  -     ----
---------------------2311104060---------------------


Welcome to Xinu!


xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```